

YURI BRAGA

Computer Engineer | Healthcare ML | Full-Stack Systems | Embedded Software

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PROFESSIONAL SUMMARY

Computer engineer with hands-on experience across healthcare ML, full-stack systems, embedded software, robotics, and electrical design. Built a production healthcare AI platform piloted with a major Brazilian healthcare payer/provider serving 10M+ covered lives, designed automated motor-controller validation systems, and led autonomous robotics/electrical architecture work across industry, research, and startup environments.

SELECTED IMPACT

10M+ Lives	0.89+ AUC	4+ Hours Saved	160+ Organizations
Healthcare AI insurer pilot	Clinical prediction models	Automated hardware validation	Full-stack platform launched

EXPERIENCE

Electrical Design Engineer, AI Fellow — Handshake AI — April 2026 - Present

- Designed, simulated, and validated analog/digital circuits using KiCad, LibrePCB, Qucs-S, and Ngspice
- Generated technical evaluations used to train AI systems on PCB design, circuit simulation, and EE reasoning
- Created prompt frameworks, documentation, and best practices enabling junior engineers to produce consistent outputs
- Reviewed deliverables and gave feedback on circuit design, simulation methodology, and troubleshooting

Co-founder & ML Lead Software Engineer — CareFuse, Inc. — Healthcare AI — March 2025 - April 2026

- Built end-to-end healthcare AI platform spanning ML models, FastAPI backend, cloud deployment, and user workflows
- Developed dual-arm prediction framework estimating patient outcomes for surgery versus conservative treatment
- Engineered patient-level feature pipelines from demographic, clinical, and patient-reported outcomes data
- Implemented nested cross-validation, Bayesian hyperparameter optimization, and calibration workflows
- Evaluated model discrimination, calibration, fairness, and clinical utility across patient cohorts
- Built SHAP-based explanation pipeline to support clinical transparency and physician trust
- Designed Dockerized deployment architecture and APIs for healthcare workflow integration
- Led pilot and customer discovery with physicians, administrators, and a major Brazilian healthcare organization

Electrical Engineering Intern — Stanley Black & Decker (DeWalt) — Towson, MD — May 2024 - August 2024

- Led design and deployment of automated motor-controller validation system, reducing manual test time by 4+ hours
- Designed test circuitry and Altium PCB integrating relay drivers, thermocouple sensing, and high-current power paths
- Developed embedded C firmware on TI MSP432 for automated sequencing, data capture, and sensor monitoring
- Wrote I2C drivers for multi-register thermocouple converter IC used for precision temperature capture
- Performed hardware bring-up, root-cause analysis, thermal validation, and reliability testing

Research Assistant — Center for Marine Autonomy & Robotics — Blacksburg, VA — August 2024 - March 2026

- Led electrical and system architecture for autonomous hydrofoil RHIB research platform
- Selected motors, batteries, microcontrollers, throttles, and power components based on system constraints
- Developed obstacle-avoidance algorithm and ROS drivers for sonar data parsing
- Integrated sensing, communications, controls, and autonomy subsystems for field robotics platform

Research Fellow — Terrestrial Robotics Engineering and Controls — Blacksburg, VA — August 2022 - May 2024

- Integrated sensing, PCB, and firmware systems for humanoid robot platform Pandora

EDUCATION

Virginia Tech — Bachelor of Science in Computer Engineering — Blacksburg, VA — GPA: 3.78/4.0

- Fralin Undergraduate Research Fellow; [VT News Spotlight Senior](#); President, ECE Ambassadors

PROJECTS

[Spurr](#) — Founder & Lead Engineer — Greater Boston — 2026 - Present

- Built and deployed full-stack platform aggregating 160+ groups and organizations events using Next.js, TypeScript, Supabase, and Vercel
- Designed PostgreSQL data models, authentication, and role-based access controls supporting complex ownership workflows
- Developed AI-assisted data ingestion, verification, moderation, and content management pipelines
- Integrated Stripe payments, email automation, analytics, deployment, stakeholder engagement, and roadmap planning

Edge AI Low-Power Drone Detection System — Union Dynamic

- Developed real-time sensing and processing pipeline for edge AI platform
- Programmed low-power real-time noise detection routine for Raspberry Pi Zero 2W

Infrared Radioteletype

- Designed radioteletype system using IR LED, Arduino, photodiode, and cascaded Butterworth filtering

LEADERSHIP

- President** — Virginia Tech ECE Ambassadors
- Led organization representing 2,000+ engineering students and coordinated outreach, recruiting, and industry initiatives
- Teaching Assistant** — Embedded Systems (ECE 2564)
- Mentored students on C programming, microcontrollers, debugging, embedded design, and hardware-software integration

SKILLS

Programming	Python, C/C++, Java, TypeScript, SQL, MATLAB
AI & Data	Scikit-learn, XGBoost, SHAP, Feature Engineering, Predictive Modeling, Statistical Analysis
Software & Cloud	FastAPI, Docker, AWS, PostgreSQL, Supabase, REST APIs, Git, Linux
Embedded & Robotics	ARM Cortex-M, RTOS, ROS2, CAN, SPI, I2C, UART, PLCs, Sensor Integration
Hardware	PCB Design, Altium Designer, Oscilloscopes, Power Electronics, Hardware Debugging
Leadership	Product Development, Technical Leadership, Cross-Functional Collaboration, Startup Operations